

(a) Retain the full prefill

One-Time SDPA Prefill



Retained cache C

Top-Budget Re-Selection



Top-Budget Re-Selection

(b) Preview the question and score content blocks



Question $\rightarrow$ Context Attention

Eager preview · per-layer hooks · reduce immediately



Block Scores

Block Scores



Top-Budget Re-Selection



(c) Build a per-query 4D additive mask

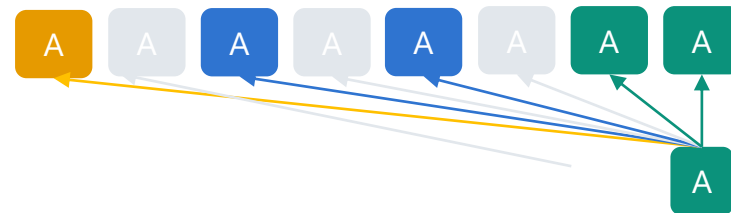
One slice per (batch, head)



Allowed Keys

(d) Decode in place under M_j

$\text{Attn}(Q, K, V, M_j) = \text{softmax}()$



$\text{Attn}(Q, K, V, M_j) = \text{softmax}()$

(e) Reversibility: discard the view, retain the memory

